2 Marks Questions:

1. What is a deterministic PDA (DPDA)?
2. Give an example of a language accepted by PDA?
3. What is the stack alphabet in PDA?
4. Define nondeterminism in PDA.
5. What is meant by transition function of PDA?
6. Can every CFL be accepted by a DPDA? Justify briefly?
7. Define equivalence of PDA and CFG?
8. What is the role of stack in PDA?
9. What is meant by CFG to PDA conversion?
10. What is meant by PDA to CFG conversion?
11. What is the start variable in CFG?
12. What is leftmost/rightmost derivation?
13. What is ambiguity in CFG?
14. Define derivation tree (parse tree)?
15. What is the significance of equivalence between CFG and PDA?
16. What is Chomsky Normal Form (CNF)?
17. What is Greibach Normal Form (GNF)?
18. What is the importance of CNF?
19. Can every CFG be converted to CNF? Justify?
20. State the pumping lemma for CFL?
21. What is the purpose of pumping lemma?
22. Can pumping lemma prove a language is CFL? Justify?
23. CFLs are closed under which operations among: Union, Intersection, Complement, Concatenation?
24. What is a decision problem?
25. Define a Turing Machine (TM)?
26. List components of TM?
27. Define recursive and recursively enumerable (RE) languages?
28. What is a universal Turing machine?
29. What is a multi-tape TM?
30. What is an undecidable problem?
31. Define halting problem?
32. What is Post Correspondence Problem (PCP)?
33. What is reduction?
34. Give an example of undecidable problem?
35. State the difference between decidable and undecidable problems?

5 Marks Questions:

1. Define PDA formally and explain its components with diagram?
2. Construct a PDA for language $L = \{ a^n b^n \mid n \geq 1 \}$?
3. Compare DPDA and NPDA?

4. Construct PDA for L = { ww^R | w ∈ {a,b}* }?
5. Convert given CFG into equivalent PDA and vice versa?
6. Explain step-by-step CFG → PDA construction?
7. Explain the steps in converting CFG to CNF?
8. State and explain pumping lemma for CFL?
9. Prove that L = { $a^n b^n c^n$ | n ≥ 1 } is not CFL?
10. Prove that L = { ww | w ∈ {a,b}* } is not CFL?
11. Explain closure properties of CFL with examples?
12. Prove CFLs are closed under union and concatenation?
13. Define TM formally and explain working with diagram?
14. Design a TM for L = { $a^n b^n$ }?
15. Explain universal Turing machine, explain halting problem?
16. Prove that halting problem is undecidable?
17. Explain Post Correspondence Problem (PCP)?
18. Explain reduction technique with example?
19. Explain undecidable problems related to Turing Machines?
20. Design a TM to check palindrome?